

Meta-Analysis for Neuroscience and Neuroepidemiology

A Practical Course from Data Extraction to Interpretation

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1 Preface

These notes are written for neuroscience, epidemiology, public health, and medical researchers who want to understand meta-analysis beyond software.

The objective is not simply to learn how to run a meta-analysis.

The objective is to understand:

1. What a meta-analysis is actually estimating.
2. What information must be extracted from papers.
3. Why effect sizes matter more than p-values.
4. How to reconstruct missing statistics.
5. How to perform a meta-analysis manually.
6. How to critically evaluate published meta-analyses.

Most confusion in meta-analysis comes from treating the process as a software exercise.

In reality, the majority of the work happens before any analysis is run.

2 Part I: Foundations

3 Systematic Review vs Meta-Analysis

Many researchers use these terms interchangeably.

They are not the same thing.

3.1 Systematic Review

A systematic review asks:

What does the literature say?

Its purpose is to identify, evaluate, and summarize all relevant evidence.

Typical steps:

1. Define a question
2. Search databases
3. Screen studies
4. Extract data
5. Assess bias
6. Summarize findings

A systematic review may never perform any statistical pooling.

3.2 Meta-Analysis

A meta-analysis asks:

If all studies are treated as pieces of one larger experiment, what is the overall effect?

Meta-analysis combines study results mathematically.

The goal is to estimate an overall effect more precisely than any individual study.

3.3 Mental Model

Systematic Review:

Finding and organizing evidence

Meta-analysis:

Combining evidence mathematically

A systematic review can exist without a meta-analysis.

A meta-analysis should rarely exist without a systematic review.

4 What Is a Meta-Analysis Estimating?

Suppose five studies investigate whether exercise improves cognition.

The results differ.

Study	Effect
A	0.3
B	0.5
C	0.2
D	0.7
E	0.4

Which one is correct?

Potentially all of them.

Each study estimates the underlying truth imperfectly.

Meta-analysis attempts to estimate the true effect.

5 The Most Important Concept

Every study must contribute two things:

1. Effect Size
2. Precision

Everything else is simply a route to obtaining those quantities.

Quantity	Question
Effect Size	How big is the effect?
Precision	How certain are we?

6 Effect Sizes

6.1 Why Effect Sizes Matter

Many researchers become overly focused on p-values.

Meta-analysis is primarily concerned with effect size.

Consider:

Study A:

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p < 0.001
Effect = 0.10
```

Study B:

```
p = 0.04
Effect = 0.80
```

Which effect is larger?

Study B.

Which result is more statistically significant?

Study A.

Meta-analysis focuses on effect magnitude.

7 Cohen's d

The most common effect size for continuous outcomes.

Used when comparing means between groups.

7.1 Formula

$$d = \frac{\bar{X}_1 - \bar{X}_2}{SD_{pooled}}$$

where

$$SD_{pooled} = \sqrt{\frac{(n_1 - 1)SD_1^2 + (n_2 - 1)SD_2^2}{n_1 + n_2 - 2}}$$

7.2 Example

Intervention:

- Mean = 25
- SD = 10

Control:

- Mean = 20
- SD = 10

Difference:

$$25 - 20 = 5$$

Therefore:

$$d = \frac{5}{10} = 0.5$$

Interpretation:

The intervention group is half a standard deviation above the control group.

8 Interpreting Cohen's d

Traditional guidelines:

d	Interpretation
0.2	Small
0.5	Moderate
0.8	Large

These are rough guidelines only.

Context matters.

In neurodegeneration research:

A d of 0.2 may be clinically important.

9 Hedges' g

Meta-analyses usually report Hedges' g rather than Cohen's d.

Why?

Cohen's d slightly overestimates effects in small studies.

Hedges' g corrects this bias.

$$g = J \times d$$

For large studies:

$$g \approx d$$

10 Standard Error

The standard error measures uncertainty.

Small SE:

More precise estimate

Large SE:

Less precise estimate

The standard error determines how much weight a study receives.

11 Confidence Intervals

A confidence interval is:

$$\text{Estimate} \pm 1.96 \times SE$$

Example:

$$d = 0.50$$

95% CI:

$$0.20, 0.80$$

Interpretation:

The data are compatible with effects between 0.20 and 0.80.

12 Recovering Standard Error from a CI

Suppose a paper reports:

$$95\%CI = (0.20, 0.80)$$

Width:

$$0.80 - 0.20 = 0.60$$

Then:

$$SE = \frac{0.60}{3.92} = 0.153$$

This formula is used constantly during extraction.

13 z Statistics

A z-statistic measures how many standard errors an estimate lies from zero.

Formula:

$$z = \frac{Estimate}{SE}$$

13.1 Example

Suppose:

$$Estimate = 0.50$$

$$SE = 0.10$$

Then:

$$z = 5$$

Interpretation:

The estimate lies five standard errors away from zero.

14 p-values

A p-value answers:

Assuming no effect exists, how surprising are these data?

A p-value does NOT tell us:

- Effect size
 - Clinical importance
 - Practical significance
-

15 Relationship Between z and p

Larger absolute z values produce smaller p-values.

Example:

z	Approximate p
1.96	0.05
2.58	0.01
3.29	0.001

16 The Meta-Analysis Detective Mindset

Every paper hides two quantities:

1. Effect size
2. Standard error

Your job is to recover them.

Different papers provide different clues.

17 Part II: Data Extraction

18 The Tier System

Not every study reports the same information.

The extraction process becomes easier if you think in tiers.

19 Tier 1: Means + SDs + Sample Sizes

Best-case scenario.

Paper reports:

Group	Mean	SD	N
Intervention	25	10	100
Control	20	10	100

19.1 Step 1

Calculate Cohen's d:

$$d = \frac{25 - 20}{10} = 0.5$$

19.2 Step 2

Approximate variance:

$$Var(d) = \frac{n_1 + n_2}{n_1 n_2} + \frac{d^2}{2(n_1 + n_2)}$$

19.3 Step 3

Calculate standard error:

$$SE = \sqrt{Var}$$

Meta-analysis ready.

20 Tier 2: Effect Size + Confidence Interval

Paper reports:

$$d = 0.5$$

95% CI:

$$0.2, 0.8$$

Recover:

$$SE = \frac{0.8 - 0.2}{3.92} \approx 0.153$$

Meta-analysis ready.

21 Tier 3: Effect Size + p-value

Paper reports:

$$d = 0.5$$

$$p = 0.01$$

Approximate:

$$z = 2.58$$

Then:

$$SE = \frac{0.5}{2.58} = 0.194$$

Meta-analysis ready.

22 Tier 4: Mean Difference + Confidence Interval

Paper reports:

$$MD = 5$$

95% CI:

$$2, 8$$

Recover:

$$SE = \frac{8 - 2}{3.92} = 1.53$$

Meta-analysis ready.

23 Tier 5: Mean Difference + p-value

Paper reports:

$$MD = 5$$

$$p = 0.01$$

Approximate:

$$z = 2.58$$

Then:

$$SE = \frac{5}{2.58} = 1.94$$

Meta-analysis ready.

24 Tier 6: Confidence Interval Only

Paper reports:

$$95\%CI = (0.2, 0.8)$$

Estimate:

$$\frac{0.2 + 0.8}{2} = 0.5$$

SE:

$$\frac{0.8 - 0.2}{3.92} = 0.153$$

Meta-analysis ready.

25 Tier 7: p-value Only

Paper reports:

$$p = 0.03$$

Nothing else.

This is usually insufficient.

Why?

Because we cannot determine:

- Direction
- Magnitude
- Practical importance

A p-value alone rarely allows inclusion.

26 Common Extraction Problems

27 Problem 1: Missing Effect Sizes

Paper reports:

Significant improvement observed ($p < 0.05$)

No means.

No effect estimate.

Need:

- Means
- SDs
- CI
- Test statistic

to reconstruct the effect.

28 Problem 2: Implausible Effect Sizes

Example:

$$d = 24$$

Almost certainly an error.

Possible causes:

- Wrong denominator
- Data-entry error
- Misinterpretation of statistics

Always return to the paper.

29 Problem 3: CI and p-value Conflict

Example:

$$95\%CI = (-0.1, 0.8)$$

and

$$p < 0.001$$

These usually cannot coexist.

This indicates extraction error.

30 Problem 4: Sample Size Mismatch

Reported:

$$N \neq n_{exp} + n_{ctrl}$$

Potential reasons:

- Attrition
- Missing data
- Cluster designs
- Subgroup analyses

Must verify.

31 Quality Control Workflow

For every study ask:

1. Is the effect size correct?
 2. Is the CI correct?
 3. Is the p-value correct?
 4. Is the sample size correct?
 5. Do all numbers make mathematical sense together?
-

32 Part III: Performing a Meta-Analysis

33 Dummy Dataset

Suppose we have four studies.

Study	Hedges g	SE
A	0.40	0.10
B	0.60	0.15
C	0.30	0.12
D	0.70	0.20

34 Step 1: Calculate Variance

$$Var = SE^2$$

Study	Variance
A	0.010
B	0.0225
C	0.0144
D	0.0400

35 Step 2: Calculate Weights

$$w_i = \frac{1}{Var_i}$$

Study	Weight
A	100
B	44.4
C	69.4
D	25

Notice:

Study A contributes much more than Study D.

36 Step 3: Pooled Effect

Fixed-effect estimate:

$$\hat{\theta} = \frac{\sum w_i y_i}{\sum w_i}$$

where:

- y_i = study effect
 - w_i = study weight
-

37 Step 4: Pooled Standard Error

$$SE_{pooled} = \sqrt{\frac{1}{\sum w_i}}$$

38 Step 5: Pooled Confidence Interval

$$\hat{\theta} \pm 1.96 \times SE_{pooled}$$

39 Heterogeneity

Studies rarely estimate exactly the same effect.

The question becomes:

How much do studies disagree?

40 I²

Interpretation:

I ²	Meaning
0%	No heterogeneity
25%	Low
50%	Moderate
75%+	High

41 Fixed vs Random Effects

41.1 Fixed Effect

Assumes:

All studies estimate the same true effect.

41.2 Random Effects

Assumes:

Different studies estimate different true effects.

This is usually more realistic in:

- Neuroscience
 - Neuroepidemiology
 - Public health
-

42 Forest Plots

A forest plot displays:

- Individual study estimates
- Individual confidence intervals
- Pooled estimate

Interpretation:

1. Direction of effect
 2. Magnitude of effect
 3. Consistency across studies
-

43 Neuroepidemiology Examples

44 Continuous Outcomes

Examples:

- MMSE score
- Cognitive score
- Brain age gap
- Memory performance

Effect measures:

- Mean Difference
 - Standardized Mean Difference
-

45 Binary Outcomes

Examples:

- Dementia incidence
- Stroke occurrence
- Depression diagnosis

Effect measures:

- Odds Ratio
 - Risk Ratio
-

46 Time-to-Event Outcomes

Examples:

- Time to dementia
- Survival after stroke

Effect measure:

- Hazard Ratio
-

47 Reading Published Meta-Analyses

Always identify:

1. Effect size
2. Confidence interval
3. Model used
4. I^2
5. Number of studies
6. Total sample size

Then ask:

- Is the effect clinically meaningful?
 - Is heterogeneity acceptable?
 - Are outliers driving results?
 - Is publication bias likely?
-

48 Final Mental Model

Every paper is a puzzle.

The paper may provide:

- Means
- SDs
- Sample sizes
- Confidence intervals
- p-values
- Odds ratios
- Hazard ratios

These are simply clues.

Your objective is always to recover:

1. How big is the effect?
2. How certain is the effect?

Once those quantities are known, the study can usually contribute to a meta-analysis.

Everything else is implementation.